

Series 2 : VG IV

THE FLATTENED GEOMETRY OF THE SPIRAL-MATRIX – FROM PLANAR SPIRAL TO SPIRAL DISC

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Date: 07.05.2026

ABSTRACT:

In VG3, we introduced the planar logarithmic spiral as a mathematical model for the dynamics of the Network. This model was useful for understanding properties such as self-similarity, continuity, and differential rotation. In VG4, we take a natural step forward: we recognize that the Network, at a cosmic scale, is not a curved line unfolding in a plane, but a **spiral disc** – a flattened structure whose thickness is negligible compared to its extent. This episode develops a simplified mathematical formalism for flattening and explains why this property is essential for cycle closure and for information conservation within the spiral-matrix.

Important: The flattening of the Network is a fundamental property present in all phases of the cosmic cycle (opening, expansion, closing). The whirling dervish metaphor used in VG3 was a first approximation that did not sufficiently suggest this flattened geometry. VG4 clarifies and integrates it into the model without invalidating previous observations.

Keywords: flattened logarithmic spiral, spiral disc, nodal thickness, 3D space, surface fractals, VG.

1. WHY VG3 WAS ONLY A FIRST STEP

In VG3, we described the spiral-matrix using the equation of a logarithmic spiral in polar coordinates:

$$r(\theta) = a \cdot e^{b\theta}$$

This equation describes a planar curve – an ideal line without thickness. It helped us understand how the Network opens and closes, how folds form, and how self-similarity is preserved. However, in reality, the Network is not an infinitely thin line. It has a **thickness** – a dimension perpendicular to the plane of the spiral, which we can call the "nodal layer."

The intuitive image we now propose is that of a vinyl record or a CD. The disc has a flat surface but also a thickness. The spiral recorded on the disc is a planar spiral, but the disc itself is a very thin three-dimensional object. Similarly, our Network is a **spiral disc** – a three-dimensional structure that develops preferentially along two dimensions, with a small but finite thickness.

2. MATHEMATICAL FORMALISM OF FLATTENING – A SIMPLIFIED SKETCH

To describe a **spiral disc**, we need three coordinates instead of two. In classical polar coordinates, we have the radius r and the angle θ to describe the spiral on the disc's surface. To describe the **thickness**, we also need a coordinate z perpendicular to the disc's plane.

We can write, for a flattened spiral, a family of equations of the form:

$$r(\theta, z) = a \cdot e^{b\theta}, \text{ with } z \text{ small}$$

In other words, the spiral unfolds on a curved surface that is **almost flat**, but with a slight undulation or variable thickness. In the ideal case of a perfectly flat disc, $z = 0$ for all points. In the real case of the Network, z is small and varies slowly.

Important note: This description is an **approximation**, just as the planar spiral was an approximation. It allows us to model phenomena at a cosmic scale without entering into details that go beyond the scope of this episode.

3. CONSEQUENCES OF FLATTENING FOR CYCLE CLOSURE

- **No gaps, but a finite thickness** – The flattened spiral contracts like a disc, not like a line that rolls up. In the closing phase, all points on the disc approach the center but retain a small, finite thickness. No gaps form between the spirals, only a compaction of the entire surface.
- **Information conservation** – Cosmic memory (the spiral-matrix) is preserved not only on a one-dimensional curve but on a **two-dimensional surface** (the disc). This means that information can be encoded and stored with greater density and stability.
- **The role of thickness in interactions** – The thickness of the disc, although small, can be the site of couplings between adjacent layers of the spiral. This explains certain resonance phenomena that we could not model with a perfectly planar spiral.

4. CONCLUSION – THE SPIRAL-MATRIX AS A SPIRAL DISC

We do not replace VG3. We complement it. VG3 gave us a first approximation – the planar logarithmic spiral. VG4 gives us a more precise approximation – the flattened logarithmic spiral, in the form of a disc. This refinement allows us to better understand how the Network closes, how information is preserved, and how adjacent layers of the spiral-matrix interact.

Important: This model of the spiral disc is only an intermediate step. In the future, we could introduce a more complex curvature or a fractal geometry of thickness. For now, it is sufficient to integrate the observation that the observable universe is nearly flat, and that our Network develops preferentially along two dimensions, while retaining a finite thickness.